Solution Manual

EE 241 - Midterm (Section 1-2), Fall 2017

Q1)

Logic function is written as:

$w = (\overline{x} \cdot \overline{y} \cdot \overline{z}) + (x \cdot \overline{y} \cdot \overline{z}) + (x \cdot y \cdot z)$ (S.O.P.)

$w = (x + y + \overline{z}) \cdot (x + \overline{y} + z) \cdot (x + \overline{y} + \overline{z}) \cdot (\overline{x} + y + \overline{z}) \cdot (\overline{x} + \overline{y} + z)$ (P.O.S.)

Q2)

```
`timescale 1ns / 1ps
module temperature(s,c,heat);
input [9:0] s;
output reg [0:0] heat;
output [5:0] c;
assign c=((s-10'd673)*3'd5)/2'd2;
always @(c)
//heat=~c[5]; // dataflow
///////////////behavioral
begin
if (c<5'd16)
heat=1'b1;
else
heat=1'b0;
end
////////////////////////
endmodule

module temperature_tb;
reg [9:0] st;
wire [5:0] ct;
wire [0:0] heat;
temperature u1(st,ct,heat);
initial
begin
st=10'd673; #10;
st=10'd675; #10;
st=10'd678; #10;
st=10'd680; #10;
st=10'd682; #10;
st=10'd685; #10;
st=10'd690; #10;
st=10'd695; #10;
st=10'd698; #10;
$stop;
end
endmodule
```

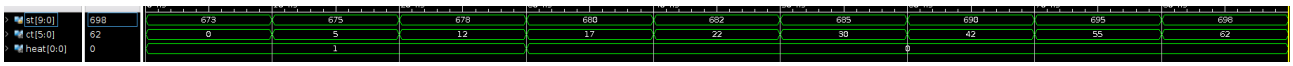


Figure 1: Q2 Simulation

Q3)

```
`timescale 1ns / 1ps
module inverse(in,out);
input [10:0] in;
output [10:0] out;
assign out=~in;
endmodule

module robot(cin,motor);
input [10:0] cin;
output reg [3:0] motor;
wire [10:0] cout;
inverse u1(cin,cout);
always @(*)
begin
case (cout)
11'b10000000000:motor=4'b0101;
11'b01000000000:motor=4'b0100;
11'b00100000000:motor=4'b0011;
11'b00010000000:motor=4'b0010;
11'b00001000000:motor=4'b0001;
11'b00000100000:motor=4'b0000;
11'b00000010000:motor=4'b1111;
11'b00000001000:motor=4'b1110;
11'b00000000100:motor=4'b1101;
11'b00000000010:motor=4'b1100;
11'b00000000001:motor=4'b1011;
default: motor=4'b0000;
endcase
end
endmodule

module robot_tb;
reg [10:0] cint;
wire [3:0] motort;
robot u1(cint,motort);
initial
begin
cint=11'b01111111111; #10;
cint=11'b10111111111; #10;
cint=11'b11011111111; #10;
cint=11'b11101111111; #10;
cint=11'b11110111111; #10;
cint=11'b11111011111; #10;
cint=11'b11111101111; #10;
cint=11'b11111110111; #10;
cint=11'b11111111011; #10;
cint=11'b11111111101; #10;
cint=11'b11111111110; #10;
$stop;
```

```
end
endmodule
```



Figure 2: Q3 Simulation